

Problem 1: Use R and B counters; repeats are allowed. A test scores one point for each matching position. Say only the total score. Use four positions. Hide a secret and make tests until you can identify it. Keep every test and score visible; add rows on scrap paper if needed. Naming a determined secret costs no extra test. Reveal and check your record.

Test	Code tested	Score
1		
2		
3		
4		
5		

Problem 2: Build each visible secret below. Test RRRR, then BRRR. Only the first test position changes. Record the scores and their difference.

Secret	RRRR score	BRRR score	Change
RBRB			
BRBR			
RRRR			
BBBB			

Problem 3: Keep a visible secret while you compare RRRR with each of BRRR, RBRR, and RRBR. Each comparison changes one position from the RRRR starting test. Record the scores for all four secrets.

Secret	RRRR	BRRR	RBRR	RRBR
RBRB				
BRBR				
RBBB				
RRRR				

Problem 4: Try those four tests on a hidden secret. Use each change from the RRRR score to find positions 1, 2, and 3. Count the Rs you have found. Subtract that count from the RRRR score to decide position 4. Record your answer, then reveal and check.

Test	Code tested	Score
1		
2		
3		
4		

My four positions and the baseline R count:

Problem 5: The first test RRRR scored 2. Build the six possible secrets below. For each proposed second test, record its score against each secret. In each row, circle the largest group of equal scores.

Test	BBRR	BRBR	BRRB	RBBR	RBRB	RRBB
BRRR						
BBRR						
BBBR						
BBBB						

Problem 6: For the second test BRRR, keep the secrets that would score 3. For the second test BBRR, keep the secrets that would score 2. Copy each group below. Try one final test against every secret in its group; record its scores.

BRRR answered 3: candidates, final test, and scores:

BBRR answered 2: candidates, final test, and scores:

Problem 7: Keep the three candidates BBRR, BRBR, and BRRB. Try the final tests below. Circle repeated scores within each row. Add a test of your own.

Final test	BBRR	BRBR	BRRB
RRRR			
RBRR			
RBBR			

Problem 8: Keep the four candidates BRBR, BRRB, RBBR, and RBRB. Try these final tests and circle repeated scores. Add a test of your own.

Final test	BRBR	BRRB	RBBR	RBRB
RRRR				
BRRR				
BBRR				

Problem 9: Take the first test BRBR and a secret of your choice. Swap R with B at positions 1 and 3 in both words, turning the test into RRRR. Record the two secret words and both scores. Repeat with a different secret.

Original secret	Score for BRBR	Changed secret	Score for RRRR